UNION & UNION All Exercise 4

Q1.

```
SELECT account_id, account_holder, NULL AS City
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder NULL AS City
FROM branch_rosebank_accounts;
```

account_id	account_holder	$ City
A001	Nomvula Dlamini	Null
A002	David Mokoena	null
A003	Lerato Sithole	Null
A004	Peter Ntosi	Null
A005	Zanele Khumalo	null
A006	Thabo Motha.	Null
~~A007~~		

Q2.

```
SELECT product_code, Product_name, NULL AS product_type
FROM savings_product
UNION
SELECT product_code, Product_name, NULL AS Product_type
FROM Current_product;
```

Product_code	product_name	product_type
SV01	Basic Savings	null
SV02	Premium Saving	null
SV03	Youth Saving	Null
SV04	Business Savings	Null
CR01	Standard Current	null
CR02	Gold Current	null
CR03	Business Current	null

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Question 3

```
SELECT  Staff_id , Staff_name, NULL AS email
FROM    retail_banking_Staff
UNION
SELECT  Staff_id , Staff_name, NULL AS email.
FROM    corporate_banking_staff ;
```

Staff_id	Staff_name	email.
S001	Mpho Ratabe	null
S002	Brian Tshabalala	null
S003	Arsha Patel	null
S004	Kabelo Moabelo	null
S005	Nandi Dube	null
S006	Sipho Khumalo	null

Question 4

```
SELECT  City_code , City_name, NULL AS region
FROM    Mobile_branch_Cities
Union
SELECT  City_code , City_name, null AS region
FROM    digital_branch_cities;
```

City_code	City_name	Region
C01	Johannesburg	null
C02	Pretoria	null
C03	Cape Town	null
C04	Durban	null
C05	Polokwane	null
C06	Port Elizabeth	null

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Question 5

```
SELECT customer_id, customer_name, NULL AS segment
FROM   push_notification_targets
UNION
SELECT customer_id, customer_name, NULL AS segment
FROM   inapp_banner_targets;
```

customer_id	customer_name	segment
C1001	Nomsa Zwane	null
C1002	Andile Buthelezi	null
C1003	Fatima Mahomed	null
C1004	Ryno van Zyl	null
C1005	Thandeka Cele	null
C1006	Samuel Nkosi	null
		~~null~~

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UNION ALL

Q6

```
SELECT txn-id, account-id, amount, CAST (NULL AS STRING)
       AS transaction-data
FROM atm01-transactions
UNION ALL
SELECT txn-id, account-id, amount, CAST (NULL AS STRING)
       AS transaction-data
FROM atm02-transactions;
```

txn-id	account-id	amount	transaction-data
T1001	A001	500-00	null
T1002	A002	1200-00	Null
T1003	A001	300-00	Null
T1004	A003	750-00	null
T1003	A001	300-00	null
T1005	A004	900-00	null
T1006	A002	450-00	null
T1007	A005	150-00	null

Q7

```
SELECT app-id, customer-id, loan-type, CAST (NULL AS INT) AS Amt-Req
FROM gauteng-loan-applications
UNION ALL
SELECT app-id, customer-id, loan-type, CAST (NULL AS INT) AS Amt-Req
FROM western-cape-loan-applications;
```

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app-id	Customer-id	loan-type	Amt-Req
LA001	C1001	Home loan	Null
LA002	C1002	Vehicle loan	Null
LA003	C1003	Personal Loan	Null
LA004	C1004	Home loan	NULL
LA003	C1003	Personal	NULL
LA005	C1005	Vehicle loan	NULL
LA006	C1006	Business	NULL
LA007	C1007	Home loan	Null

CAST (NULL AS STRING) AS logged_date

Q6

```
SELECT complaint-id, customer-id, CAST(NULL AS STRING) AS category,
FROM email-complaints
UNION ALL
SELECT complaint-id, customer-id, CAST(NULL AS STRING) AS CATEGORY, CAST(NULL AS STRING)
FROM app-complaints ;
```

AS logged_date

complaint-id	customer-id	category	logged_date
EC001	C2001	Null	NULL
EC002	C2002	Null	Null
EC003	C2003	Null	Null
EC004	C2004	Null	Null
AC001	C2005	Null	Null
AC002	C2001	Null	Null
AC003	C2006	Null	NULL
AC004	C2007	Null	NULL

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Q9

```
SELECT payment-id, account-id, amount
FROM april-payments
UNION ALL
SELECT payment-id, account-id, amount
FROM may-payments
```

payment-id	account-id	amount
Pay 001	A001	12 500
Pay 002	A002	4800
Pay 003	A003	9200
pay004	A004	3300
Pay 005	A001	12500
Pay 006	A005	7600
pay 007	A002	5100
Pay 009	A006	2800

Q10

```
SELECT entry-id, account-id, entry-type, CAST (NULL AS INT)
        AS amount, CAST (NULL AS STRING) AS entry-data
FROM debit-entries
UNION ALL
SELECT entry-id, account-id, entry-type, CAST (NULL AS INT)
        AS amount, CAST (NULL AS STRING) AS entry-data
FROM credit_entries;
```

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Entry_id	account_id	entry_type	amount	entry_date
DR001	A001	Debit	NULL	NULL
DR002	A002	Debit	NULL	NULL
DR003	A003	Debit	NULL	NULL
DR004	A004	Debit	NULL	NULL
CR001	A001	Credit	NULL	NULL
CR002	A005	Credit	NULL	NULL
CR003	A002	Credit	NULL	NULL
CR004	A006	Credit	NULL	NULL

3. Bonus Questions

B1. They should use UNION because it combines the two tables (January and February)
It also removes duplicates

B.2. use UNION ALL
-It will count all individual transaction
-keeps all rows including duplicates.
-It is faster because it does not remove duplicates.

B3. The two query has different number of colums the first one has two columns and the second one has three columns.
when using Union the select statements from both tables must have the same number of columns.

```
SELECT account_id, account_holder, city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, city
FROM branch_rosebank_accounts;
```

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